



Home



My Network



Jobs



Messaging



Notifications



Me ▼

Public profile settings

You control your profile and can limit what may be shown in search tools and other off-LinkedIn services. Viewers who aren't signed in to LinkedIn may see all or some portions of the profile view displayed below.



Richard Tseng

Senior Data Engineer specializing in Data Analysis and Engineering at American Express

Cambridge, Massachusetts, United States

982 followers · 500+ connections

Join to view profile



American Express



Columbia University in the City
of New York



[Portfolio](#) ↗



Home



My Network



Jobs



Messaging



Notifications



Me ▼

American Express leverages over three years of expertise in data engineering to design and optimize scalable, real-time data pipelines. With a focus on ETL architecture, Python, and Google Cloud Platform, the team supports critical business objectives by delivering efficient, reliable, and robust data solutions.

By fostering collaboration with cross-functional teams, the organization integrates dynamic feedback mechanisms and advanced optimization techniques to ensure timely delivery of impactful projects. This commitment empowers data-driven decision-making and drives innovation to meet business needs effectively.

Activity

982 followers

Posts

Comments

Reactions

Preview only

 This is a sample view to give you an idea of how your Activity section will look. The live version may appear differently.



Home



My Network



Jobs



Messaging



Notifications



Me ▼

Experience



American Express

3 years 9 months

Senior Data Engineer

Dec 2023 - Present · 2 years 4 months

- Led the design and optimization of real-time data pipelines to ensure efficiency and scalability
- Provided valuable feedback in design reviews, enhancing ETL architectures
- Communicated updates and deadlines effectively, showcasing adept time management skills for timely project delivery.
- Consolidated datasets for commercial products, card transactions, and firmographics to streamline product analytics
- Supported cross-product spend behavior insights to support data-driven...

Show more ▼

Data Engineer

Jul 2022 - Dec 2023 · 1 year 6 months

United States



in4mation insights

1 year 4 months

Senior Analyst, Retail Marketing Science

Mar 2022 - Jun 2022 · 4 months

Greater Boston

- built dynamic price elasticity model to track the change of customer attitude toward price change over time

Analyst, Retail Marketing Science

Mar 2021 - Mar 2022 · 1 year 1 month



Home



My Network



Jobs



Messaging



Notifications



Me ▼

retention analysis

- Data Science Research

- Designed an AB test (design of experiment - mixed-level fractional factorial design) and reduced 66% of runs by balancing resources and interpretability
- Implemented simulation to figure out the confounding level for interactions in Resolution V design (Experimental design)
- Researched the mechanism and implementation of multi-objective heuristic...

[Show more ▼](#)

Finance Data Science Intern

American Express

Jul 2020 - Aug 2020 · 2 months

New York, United States

Forecast Change in Ultimate Redemption Rate for Membership Rewards Program

- Yielded up to \$21M forecasting accuracy in P&L planning and identified 6 significant model predictors
- Explored XGBoost, lightGBM, and ML to lower 50% variance in credit card reward redemption rate forecast
- Constructed modeling pipeline which explored different Sklearn models and reduced experimentation time by 50%
- Visualized past trend and model predictions to pinpoint critical effects and provide...

[Show more ▼](#)

Python Developer / Quantitative Analyst

MacroMicro

Oct 2018 - Jun 2019 · 9 months

Taipei City, Taiwan

- Conceptualized and built a backtesting system specifically for macroeconomic indicators (Python, PHP, MariaDB)
- Developed backend APIs for web service to backtest trading strategies over financial products



Home



My Network



Jobs



Messaging



Notifications



Me ▼

- Organized a web scraping system that reduced the error rate by 70% using Python (BeautifulSoup4)
- Redesigned recommendation system for charts and articles using SVD, tf-idf and cosine...

[Show more ▼](#)
**Junior Data Scientist**

MoBagel

Jul 2017 - Jul 2018 · 1 year 1 month

Taipei City, Taiwan

- Built a fully automated data analysis platform by using scala and python within three months
- Implemented unit tests of the platform with >90% coverage rate
- Applied machine learning and statistical method on manufacturing, retail, human resource, and telecommunication industries to enhance performance and efficiency
- Improved accuracy or decreased errors for 4 data analysis projects by at least 15%
- Ran Scrum to develop our product
- Designed hypothesis tests (Wilcoxon...

[Show more ▼](#)
**Speaker**

PyCon Taiwan

Jun 2018 - Jun 2018 · 1 month

Taipei City, Taiwan

- Delivered a speech "Machine learning system with high scalability" with Kevin Shyu
- Was selected as final 31 presenters

Python (Pandas, Numpy, Plotly, Sklearn), Docker, MongoDB (NoSQL), SOLID OOP

**Trainee**

Sim M Cloud (closed)

Jul 2015 - Aug 2015 · 2 months

Hsinchu County/City, Taiwan



Home



My Network



Jobs



Messaging



Notifications



Me ▼

- Wrote several investment articles that introduce common investing strategies and prior knowledge
- Tested the usefulness of common investing strategies(technical analysis and fundamental analysis)

Education



Columbia University in the City of New York

Master of Science - MS · Operations Research

2019 - 2020

Activities and Societies: Taiwanese Student Association (TSA), Federation of Taiwanese Student Association in NY (FTSANY)

- Enroll in 2019 Fall
- Plan to finish "Machine Learning and Artificial Intelligence" concentration
- Courses: Optimization (prof. Christian Kroer), Stochastic Models (prof. David Yao), Algorithms (prof. Clifford Stein), Personalization: Recommendation Theory and Applications (prof. Brett Vintch, Brianne Cortese), Computational Discrete Optimization (prof. Yuri Faenza), Algorithm Trading (prof. Kenneth Gleason), AI & OR at Scale in the cloud (prof. Pierre-Hadrien Arnoux), Database (prof...

Show more ▼



National Cheng Kung University

Bachelor of Business Administration - BBA · Statistics · GPA 3.86/4.0 (4.18/4.3), ranked second among my peers

2013 - 2017

Activities and Societies: academic club - ICC, badminton

Projects:

- Classification based on density ratio estimation with nonlinear subspace projection
- Analyzing bipolar disorder medical reports provided by National Taiwan University Hospital
- Assisting Council of Agriculture to improve the accuracy and efficiency of sampling survey
- Comparison between Kernel SVM and Neural Network on urban land cover



Home



My Network



Jobs



Messaging



Notifications



Me ▼

• Finding the relation...

Show more ▼

Volunteer Experience



Reviewer

Taiwan Institute for Sustainable Energy

Feb 2016 - Sep 2016 · 8 months

Economic Empowerment

- give guidance in Corporate Social Responsibility for corporations
- review their reports



Marketing Financial Analyst

Inspiring

Sep 2015 - Jul 2016 · 11 months

Social Services

we helped Yujing, Tainan to sale their products and return profits to local children education
analyzed Facebook visitors and financial reports to make marketing and financial strategies

Licenses & Certifications



AI Engineer Agentic Track: The Complete Agent & MCP Course

Udemy

Issued Mar 2026

Credential ID UC-4cc2e7c5-4d2a-4e51-97fc-39ea823509a8

[See credential](#) ↗



Home



My Network



Jobs



Messaging



Notifications



Me ▼

Udemy

Issued Feb 2026

Credential ID UC-9bfff6fc-c0b6-49d9-aac8-9cd5b0da6650

[See credential](#) **From AI Strategist to AI Leader**

Udemy

Issued Feb 2026

Credential ID UC-e8aaada9-ccd8-4c5b-96d4-e8672a0a265e

[See credential](#) **AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents**

Udemy

Issued Jan 2026

Credential ID UC-6f536f8d-b1a2-407e-b4a0-0ed69777cd79

[See credential](#) **AI Agents Intensive Course with Google**

Kaggle

Issued Dec 2025

[See credential](#) **ODSC 2025: 5-Week Spring AI Bootcamp**

Open Data Science Conference (ODSC)

Issued May 2025

**Prompt Design in Vertex AI Skill Badge**

Google

Issued Mar 2025

[See credential](#) **Data Engineering Bootcamp**

DataExpert.io

Issued Jan 2025



Home



My Network



Jobs



Messaging



Notifications



Me ▼



Associate Cloud Engineer

Google

Issued Feb 2022

Credential ID 47066548

[See credential](#)



Ultimate Google Certified Associate Cloud Engineer 2020

Udemy

Issued Feb 2022

Credential ID UC-69903e45-2bd5-4998-9a29-a7779a08f492

[See credential](#)



Spatial Data Science and Applications

Coursera

Issued Jul 2021

Credential ID C8QRJ8QJ69P6

[See credential](#)



How Google does Machine Learning

Coursera

Issued Sep 2019

Credential ID FU7PXDHALHCA

[See credential](#)



Launching into Machine Learning

Coursera

Issued Sep 2019

Credential ID Z74KRC434MLA

[See credential](#)



Git Started with GitHub

Udemy

Issued Aug 2018



Home



My Network



Jobs



Messaging



Notifications



Me ▼



Learning Python for Data Analysis and Visualization

Udemy

Issued May 2017

Credential ID UC-WSA64X6B

[See credential](#)

Courses

Advanced Biostatistics

STAT 6054

Design of Experiments

STAT 3016

Finanical Statistics

STAT 3009

Game Theory

ECON 4002

Intelligence Manufacturing Systems

IMIS 7232

Investment Analysis

STAT 3010

Machine Learning

STAT 3024

Market Survey

STAT 4021



Home



My Network



Jobs



Messaging



Notifications



Me ▼

Nonparametric Statistics

STAT 3003

Statistical Consulting

STAT 4008

The Analysis of Categorical Data

STAT 3012

Time Series Analysis

STAT 3022

Algorithms

CSOR 4231

Optimization

IEOR 4004

Personalization: Recommendation System

IEOR 4571

Stochastic Model

IEOR 4106

Projects

Business Development Agentic AI Workflow Automation Project

Feb 2026 - Feb 2026

- n8n, RAG, Voice AI

- Designed and deployed autonomous AI agents using n8n to plan, reason, and execute multi-step business workflows across Google Sheets, Gmail, Slack, and APIs with conditional logic and webhooks.



Home



My Network



Jobs



Messaging



Notifications



Me ▼

structured, reliable outputs.

- Implemented AI voice agents integrated with ElevenLabs and Twilio...

Show more ▼

ReguAI: AI Agent for Systematical Compliance Reporting for Highly Regulated Industries

Dec 2025 - Dec 2025

- Architected ReguAI, an end-to-end agent-based AI compliance reporting system using Google ADK
- Constructed complex multi-agent orchestration and evaluation layers for consistent and fully traceable compliant outputs
- Applied sequential, parallel, and loop workflows with built-in evaluation to deliver audit-ready and reproducible reports

US salary AI agent

May 2025 - Jun 2025

1. Cleaned and normalized over 1 million+ raw H-1B salary records from Department of Labor datasets, enabling accurate downstream analytics
2. Designed and implemented a natural language to SQL pipeline using OpenAI's GPT models, supporting free-form user queries in real time
3. Built a dynamic data visualization engine that auto-selects chart types (via Plotly) based on query context, improving data comprehension
4. Deployed full-stack AI service on Google Cloud Platform (GCP) using...

Show more ▼

TravelBeing

Feb 2024 - Jan 2025

1. Business Proposal Development: Gained experience in crafting comprehensive business proposals that effectively communicate project goals, strategies, and expected outcomes, ensuring alignment with stakeholder expectations.
2. WordPress Webpage Setup: Developed proficiency in setting up and customizing WordPress websites, enhancing user experience through tailored themes, plugins, and SEO optimization.
3. Google Tag Implementation: Acquired skills in implementing Google Tag...



Home



My Network



Jobs



Messaging



Notifications



Me ▼

Simulation in Inventory Management and Dynamic Pricing for Fresh Food Business

Oct 2020 - Dec 2020

- Mimicked fresh food daily sells and largely enhanced profit by 11 times by selecting the best pricing strategy
- Identified an average 3% yearly profit decrease by natural disaster (Hurricane) for further business planning

[See project](#) ↗

Improvement of Tabu Search for Portfolio Optimization

Apr 2020 - Jun 2020

- Strategically selected 30 stocks from S&P 500 stocks from days to within 5 minutes
- Modified Tabu Search and increased ROI by 46% to 61% compared with the performance of the S&P 500

Lyrics to Music Alignment by Using Kaldi

Apr 2020 - Jun 2020

- Modeled on 2300+ Ted talk with randomly assigned background music
- Built Monophone and Triphone models by using Gaussian Mixture Model and Expectation-Maximization Algorithm

Building a Yelp Recommendation Service by segmenting users and stores

Oct 2019 - Dec 2019

- Cross Modeled on 6.7M reviews over 1.6M users across 192+ thousand stores
- Reduced MSE by 65% by applying Factorization Machine with text mining and ensemble methods

Building an Offline Movie Recommendation Service

Sep 2019 - Nov 2019

- Modeled on 20 million rating histories over 140 thousand users across 28 thousand movies
- Manipulated Spark RDD, numpy CSR matrix and pandas DataFrame for preprocessing
- Built a nonnegative ALS model for all users that decreased RMSE by 21% (compared with baseline)
- Reduced RMSE by 27% for recommendation to active users

(Python, Spark, SQL, SkLearn)



Home



My Network



Jobs



Messaging



Notifications



Me ▼

Analyzing Bipolar Disorder Medical records

Sep 2016 - Feb 2017

Analyzing bipolar disorder data provided by National Taiwan University Hospital

Improve the accuracy and efficiency of sampling survey

Sep 2015 - Feb 2017

Assisting Council of Agriculture to improve the accuracy and efficiency of sampling survey

Honors & Awards

Microsoft GCR Startup Hackfest - Smart Retail

Microsoft Taiwan, Coca Cola

Jan 2018

Microsoft had invited Coca-Cola, AB InBev, launching this Startup Hackfest which is a hacking contest to address retail business challenges through technical discussion and coding sessions.

Our company won the first prize in Taiwan, and entered Microsoft Accelerator Beijing

TiC100 Social Enterprise Competition

Tic 100

Jul 2016

starting Inspiring after this competition
we received Community Unity Prize

Languages

Chinese

Native or bilingual proficiency

Professional working proficiency

More activity by Richard



Good representation 🙌🙌
Liked by Richard Tseng

... for the solo founder, and one person, every department, shipping at startup speed.

Departments	
Department	Mission
Engineering	Build the product – frontend, backend, AI, infra, prototyping
Product	Shape what gets built – trends, feedback, priorities
Marketing	Get the word out – content, social, growth, community, ASO
Design	Make it right – UI, UX, brand
Project Mgmt	Keep it moving – shipping cadence, experiments
Operations	Keep it running – support, analytics, infra, legal, finance

Well damn. I officially built my first one. A startup with zero employees. 6 months ago I...
Liked by Richard Tseng



Home



My Network



Jobs



Messaging



Notifications



Me ▼

```
reports\retail_data_v3\summary_statistics.csv
reports\retail_data_v3\schema.json
reports\retail_data_v3\missingness.csv
reports\retail_data_v3\correlation.csv
reports\retail_data_v3\duplicates.csv
reports\retail_data_v3\outliers.csv
reports\retail_data_v3\data_quality.json

Generating plots ...
reports\retail_data_v3\plots\distribution_histogram.png
reports\retail_data_v3\plots\correlation_heatmap.png
reports\retail_data_v3\plots\missingness_bar.png
reports\retail_data_v3\plots\box_plot.png

Done. 2/8 columns have missing data (136,604 total missing values).
Suspicious values treated as missing:
Description: 11 values (e.g. '?lost', 'Missing', '?missing')
Quality issues: 0 critical, 5 warnings, 0 info.
WARN [Description]: Column 'Description' contains leading/trailing whitespace in text values.
WARN [CustomerID]: Column 'CustomerID' is 24.93% missing (135080/541909 values).
WARN [InvoiceNo]: Column 'InvoiceNo' is text but 08.20% of non-empty values are numeric-passable
```

AI coding tools are everywhere right now. It is genuinely powerful, but using it well is its...

Liked by Richard Tseng

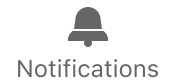
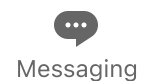


Another awesome role at Suntory. Sarah Tomasaitis is a phenomenal leader and this role i...

Liked by Richard Tseng



Myself and Holly Heglin are growing our AI team at Jarvis Consulting Group ! I'm hiring for...



|

...

Liked by Richard Tseng



If your company is not keeping up with AI, this is why: You don't have enough Jacks (aka...

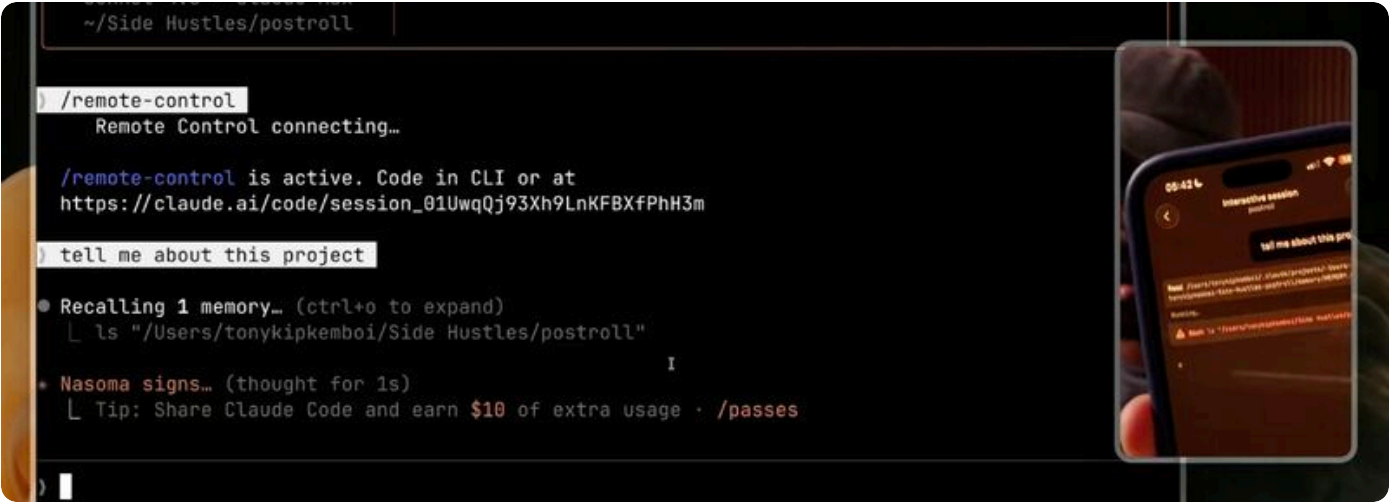
Liked by Richard Tseng



11	columns:				
12	- name: TRANSACTION_ID	QUANTITY	No invalid values	level: fail must be: 0	❌ FAILED
13	data_type: number				
14	checks:				
15	- duplicate:				
16	- missing:				
17	- name: STORE_ID	SALES_AMOUNT	No invalid values	level: fail must be: 0	✅ PASSED
18	data_type: number				
19	- name: PRODUCT_ID				
20	data_type: number	TRANSACTION_ID	No duplicate values	level: fail must be: 0	❌ FAILED
21	checks:				
22	- missing:				
23	- name: CUSTOMER_ID				
24	data_type: number		No missing values	level: fail must be: 0	✅ PASSED
25	checks:				
26	- missing:				
27	- name: DATE				
28	data_type: date				
29	- name: QUANTITY				
30	data_type: number	[dataset-level]	Data is fresh	level: fail must be less than: 1	❌ FAILED
31	checks:				
32	- invalid:				

I am a data architect. What use are data contracts to me? If you design data platforms at...

Liked by Richard Tseng



are we slowly entering the era of "continous work"? 🤖 Anthropic released a new feature...

Liked by Richard Tseng



Edit your custom URL

Personalize the URL for your profile.

www.linkedin.com/in/yi-ping-tseng



Edit content

This is your public profile. To edit its sections, update your profile.

Edit visibility

You can set limits on which profile fields are eligible for display to people who aren't signed in to LinkedIn. Those viewing your profile in search tools, profile badges, or other permitted services like Outlook, may see all or some of the profile fields you have enabled below.

[Learn more](#)

Your profile's public visibility

On

Basic (required)

☒ Name, number of connections, number of followers, and region

Profile photo

☐ Only 1st-degree connections

LinkedIn members directly connected to you.

☐ Your network

Your connections, up to three degrees away from you.

☐ All LinkedIn members

☒ Public

All LinkedIn members on or off LinkedIn. Your content could be visible in search results (Google, Bing, etc.).

Background photo

Allow

Headline

Allow

Websites

Allow

Summary

Allow

Articles & activity

Allow

Current experience

Allow

Details

Allow

Details

Allow

Education

Allow

Details

Allow

Volunteer experiences

Allow

Certifications

Allow

Courses

Allow

Projects

Allow

Honors and awards

Allow

Languages

Allow

 Public profile badge

Promote your profile by adding a badge to your blog, online resume, or website.

Create a badge

LinkedIn © 2026

About

Accessibility

User Agreement

Privacy Policy

Your California Privacy Choices

Cookie Policy

Copyright Policy

Brand Policy

Community Guidelines

Help Center

Settings

Language ▼